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Effects of captopril treatment versus placebo on renal function in type 2 diabetic patients with microalbuminuria: a long-term study

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Abstract. We evaluated the renal effect of long-term antihypertensive treatment (12 months) with the angiotensin-converting enzyme inhibitor captopril compared to placebo in 15 type 2 diabetic patients with microalbuminuria. The patients were randomly allocated to captopril ($n=9$) or placebo ($n=6$). After 1-year therapy no significant decrease in blood pressure was demonstrated with captopril ($139\pm17/80\pm9$ versus $138\pm13/76\pm6$ mmHg) or placebo ($138\pm9/75\pm6$ versus $135\pm14/79\pm10$ mmHg). Only in a small hypertensive subgroup ($n=4$) treated with captopril did we find a significant reduction in blood pressure ($154\pm2/88\pm1$ versus $142\pm7/78\pm5$ mmHg, $P<0.05$). The urinary albumin excretion rate did not change significantly either in the captopril group (95.6 mg/24 h, 25th percentile 138.4, 75th percentile 25.1; versus 127.8 mg/24 h, 25th percentile 29.3, 75th percentile 222) or in the placebo group (99.2 mg/24 h, 25th percentile 58.5, 75th percentile 125.8; versus 120.9 mg/24 h, 25th percentile 62.1, 75th percentile 179.7). There were also no alterations in renal blood flow or filtration rate. In the hypertensive subgroup treated with captopril a reduction in urinary albumin excretion rate after 3 and 6 months of treatment was observed (captopril 73.4 versus 24 and 41 mg/24 h, $P<0.05$), but not after 12 months. Triglyceride and cholesterol levels remained constant before and after treatment while glycosylated hemoglobin decreased significantly after 12 months captopril (7.8 ± 0.9 versus 6.9 ± 0.7 mg%, $P<0.03$). We conclude that in patients with type 2 diabetes with microalbuminuria angiotensin-converting enzyme inhibitors may have protective renal effects in so far as a lack of increase in urinary albumin excretion is equivalent to low progression in renal disease.

Key words: Angiotensin-converting enzyme inhibitor – Type 2 diabetes mellitus – Microalbuminuria

Abbreviations: ACE = angiotensin-converting enzyme; GFR = glomerular filtration rate; HbA_{1c} = glycosylated hemoglobin; RPF = renal plasma flow; UAER = urinary albumin excretion rate

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Diabetic nephropathy develops in about 40% of patients with type 1 diabetes mellitus with a diabetes duration of more than 20 years [8, 14, 36] and in 15%–60% of patients with type 2 diabetes [7, 27]. Microalbuminuria in type 2 diabetes is associated with a high mortality from cardiovascular events [17, 33, 34]. Overt proteinuria is preceded by subclinical elevation in urinary albumin excretion, which has been termed microalbuminuria [24]. Microalbuminuria per se is associated with several risk factors for renal and atherosclerotic disease, such as increased levels of total and low-density lipoprotein cholesterol [9, 35], triglycerides, plasma fibrinogen and most strikingly elevated arterial blood pressure [20, 25].

Many studies in type 1 diabetes have shown the efficacy of early intervention in arresting or delaying the progression from microalbuminuria to overt nephropathy, which occurs in untreated patients after a mean period of 7 years [28, 29]. Apart from improvements in metabolic control [13] and protein restriction [10, 11], the main focus has been directed at reducing the elevated arterial blood pressure. Among the various groups of antihypertensive drugs angiotensin-converting enzyme (ACE) inhibitors and Ca antagonists do not exert adverse effects on insulin sensitivity or metabolic

control [30] and are therefore most commonly studied. ACE inhibitors have been shown to control blood pressure adequately and decrease proteinuria in patients with overt diabetic nephropathy [5, 18]; even in normotensive patients a reduction in microalbuminuria by ACE inhibitors has been demonstrated [19]. In type 2 diabetic patients, however, data concerning the long-term effects of ACE inhibitors on microalbuminuria are limited [6, 22, 32]. Most studies have included only few patients and for a relatively short period [1, 12, 19, 23].

This study was designed to compare the long-term effects of ACE inhibitor captopril to placebo on the urinary albumin excretion rate (UAER) and progression of renal disease in patients with type 2 diabetes and microalbuminuria.

Materials and methods

A total of 20 patients with type 2 diabetes and microalbuminuria began the study, 10 of whom were randomly assigned to captopril and 10 to placebo. Of these, 15 were observed for 12 months (captopril, 5 men and 4 women; placebo, 2 men and 4 women). The reasons for these five not completing the study were: one experienced side effects during treatment with captopril (cough and low blood pressure), three did not want to continue the follow-up visits regularly, and one moved away. The subjects were selected on the basis of age, urinary albumin excretion with negative urine cultures, normal creatinine levels, and the presence of diabetic retinopathy. The clinical characteristics of the two groups are given in Table1. No significant differences were seen between the groups regarding sex, age, or duration of diabetes. Patients were comparable in their baseline characteristics. Each gave informed consent, and the study was approved by the local ethics committee and performed according to the Declaration of Helsinki.

Microalbuminuria is diagnosed on the basis of increased UAER, ranging from 30 to 300 mg/24 h measured in three or more 24 h urine collections. All other possible causes, such as urinary tract infection, were excluded (urine samples were checked by bacterial cultures). Patients with heart disease, pregnancy, or impaired renal function were excluded (serum creatinine < 1.4 mg/dl). Diabetic retinopathy was evident in every patient before starting therapy examined by fundus photography (11 patients with nonproliferative and four with proliferative retinopathy).

Table 1. Clinical characteristics of 15 type 2 diabetic patients with microalbuminuria

	Captopril ($n=9$)	Placebo ($n=6$)
Age (years)	64 ± 8.2	63.8 ± 1.1
Duration (years)	12.9 ± 3.4	11.8 ± 3.9
Body mass index	26.7 ± 2.8	28.2 ± 2.6
Creatinine (mg/dl)	0.96 ± 0.2	0.94 ± 0.3

Four patients with repeatedly elevated arterial blood pressure measurements (systolic blood pressure > 140–200 mmHg, diastolic blood pressure > 85–115 mmHg) were defined as hypertensives (one man, three women). The mean duration of diabetes in the captopril group was 13 years. One patient was being treated with diet alone, three with oral hypoglycemic agents (glibenclamide therapy) and diet, and five with insulin and diet. In the placebo group the mean duration of diabetes was 12 years and all patients except one (glibenclamide and metformin therapy) were being treated with insulin and diet. To maintain a diet consisting of 50% carbohydrate, 20% protein, and 30% fat the patients were instructed by our diet assistent before the beginning of treatment and every 6 weeks thereafter. The daily dosage of in the captopril group ($n=9$) was 37.5 mg (3×12.5 mg).

Blood pressure measurement was made with a stardard sphygmomanometer (cuff 25×12 cm) with the patient seated for the preceding 15 min. The mean values of three blood pressure measurements were used for statistical evaluations. In our laboratory urine albumin was determined using a commercial radioimmunoassay (Pharmacia, Uppsala, Sweden), the average of three UAER values of 24-h urine samples was calculated for further analysis. Every 3 months we supervised blood pressure, routine blood chemistry including blood glucose, lipids (triglycerides, cholesterol), serum creatinine and electrolytes, and glycosylated hemoglobin A1 (HbA_{1c}). UAER was controlled at the beginning and after 3, 6, and 12 months. These parameters were measured with an American Monitor Parallel Analyzer (Richmond, USA), HbA_{1c} was measured by fast liquid pressure chromatography [16]. Glomerular filtration rate (GFR) and renal plasma flow (RPF) were measured by clearance of ^{51}Cr-labeled EDTA and ^{123}I-labeled orthohippurate (plasma clearance or single injection clearance) [3, 4, 31]. After a single intravenous injection of 100 μCi ^{51}Cr-labeled EDTA and ^{123}I-labeled orthohippurate in the right arm, blood samples were taken from the other arm, and plasma was separated. The decrease after 20, 40, 60, 90, 120, 180, and 240 min was determined. Clearance was calculated and corrected for body surface area of 1.73 m^2.

Statistical analysis used Student's paired t tests and variance analyses. Values are expressed as means and standard error of the mean or standard deviation or as medians and quartiles. A P value of 0.05 or lower was considered statistically significant.

Results

After 12 months we observed no effect on blood pressure either in the captopril group (139±17/80±9 versus 138±13/76±6 mmHg) or in the placebo group (138±9/75±6 versus 135±14/79±10 mmHg; Table 2). In only a small subgroup of captopril-treated patients with initially higher blood pressure (the hypertensive subgroup) was a significant decrease found in systolic and diastolic blood pressures (154±2/88±1 versus 142±7/78±5 mmHg, $P<0.05$; Table 3).

Concerning metabolic parameters we found a significant decrease in HbA_{1c} after 12 months of antihypertensive treatment (7.8±0.9 versus 6.9±0.7 mg%, $P<0.03$) whereas in the placebo group a slight increase was observed (8.2±1.3 versus 9.3±1.6 mg%, n.s.; Fig. 1).

In neither group was a significant change observed in cholesterol level before and after therapy (captopril, 241.8±50.9 versus 232.5±56.6 mg/dl, n.s.; placebo 230±60.4 versus 237±47.8 mg/dl, n.s.). There was also no change in triglycerides (captopril, 171.3±57 versus 173.3±83.6 mg/dl, n.s.; placebo, 231±128.6 versus 278.2±192.6 mg/dl, n.s.).

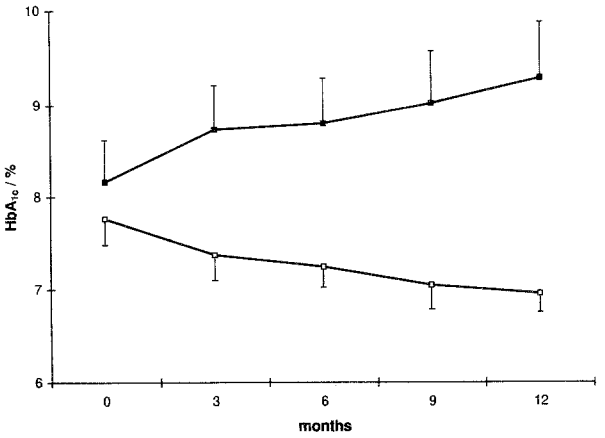

Fig. 1. HbA_{1c} in type 2 diabetic patients ($n=15$) controlled every 3 months. □, Captopril; ■, placebo. Means ± SEM)

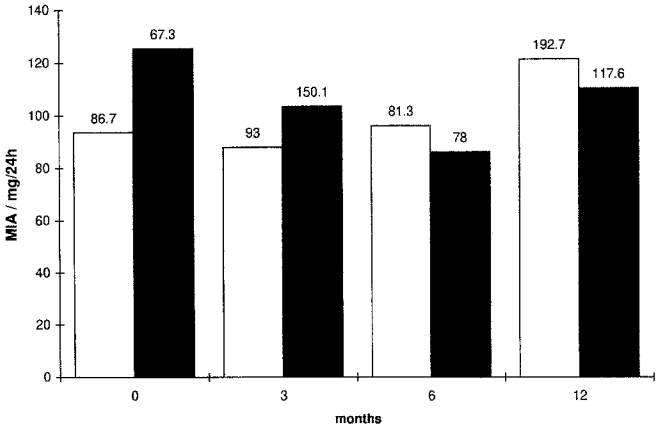

Fig. 2. Microalbuminuria (*MIA*) in type 2 diabetic patients treated with captopril ($n=9$) and placebo ($n=6$). Median and range (75th–25th percentiles); □, captopril; ■, placebo

UAER did not change significantly in patients treated with captopril compared to those treated with placebo (Fig. 2). Only in the hypertensive subgroup was a significant decrease observed in UAER after 3 and 6 months (74.4 versus 24 and 41 mg/24 h, $P<0.05$); however, after 12 months the initial level was reached again (Fig. 3).

Antihypertensive treatment with captopril showed similar effects on GFR and RPF compared to placebo. We found no changes in GFR during captopril (97.2±24.4 versus 94.8±25.9 ml/min × 1.73 m^2, n.s.) or placebo treatment (94.4±18.9 versus 87.1±22.4 mg/min × 1.73 m^2 n.s.; Fig. 4), nor in RPF (captopril, 436.6±186.1 versus 435.1±195.2 mg/min × 1.73 m^2 n.s.; placebo 456.8±219.6 versus 447± 75.8 mg/min × 1.73 m^2 n.s.; Fig. 5).

Table 2. Systolic and diastolic blood pressure (mmHg) in 15 type 2 diabetic patients with microalbuminuria treated with captopril or placebo before and after 12 month treatment

	Systolic	Diastolic
Captopril		
Before	139±17	80±9
After	138±13	76±6
Placebo		
Before	138±9	75±6
After	135±14	79±10

Table 3. Systolic and diastolic blood pressure (mmHg) in four hypertensive type 2 diabetic patients with microalbuminuria treated with captopril before and after 12-month antihypertensive treatment

	Systolic	Diastolic
Before	154±2	88±1
After	142±7*	78±5*

* $P<0.05$

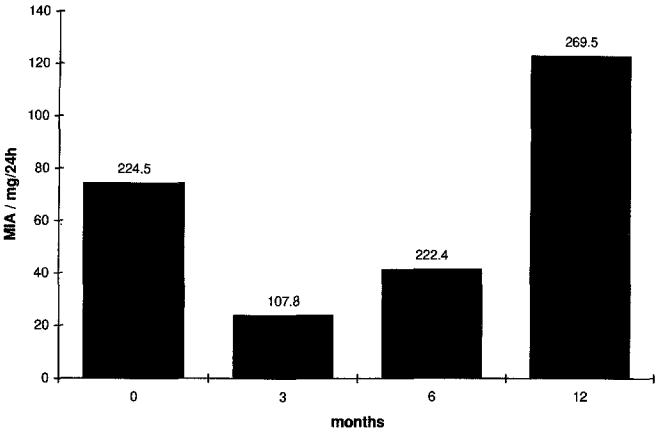

Fig. 3. Microalbuminuria (*MIA*) in hypertensive type 2 diabetic patients treated with captopril ($n=4$). Median and range (75th–25th percentiles)

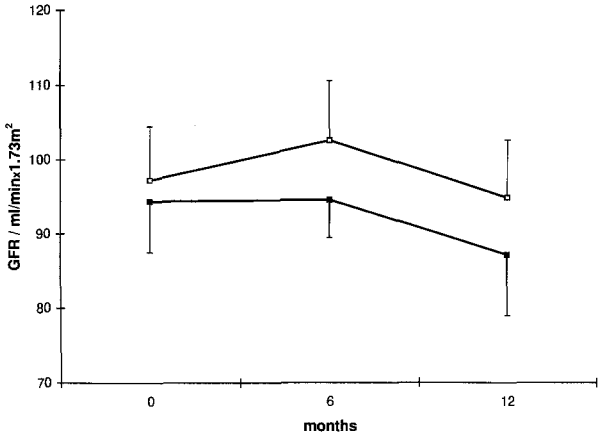

Fig. 4. Glomerular filtration rate in type 2 diabetic patients ($n=15$) at the beginning and after 6 and 12 months. □, Captopril; ■, placebo. Means ± SEM

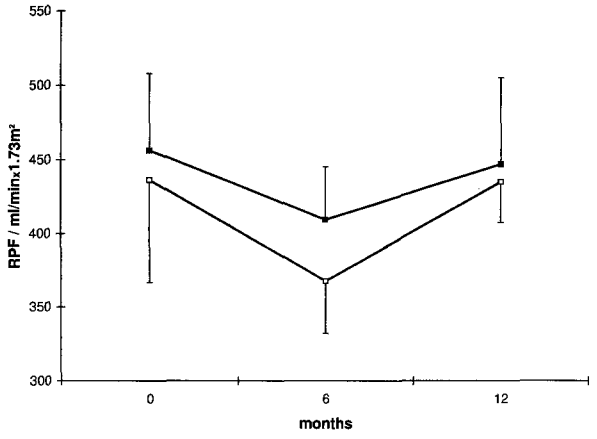

Fig. 5. Renal plasma flow in type 2 diabetic patients ($n=15$) at the beginning and after 6 and 12 months. □, captopril; ■, placebo. Means ± SEM

Discussion

The main finding of our study is that captopril treatment for 12 months does not reduce UAER in type 2 diabetic patients. The significant reduction in UAER in a small hypertensive subgroup after 3 and 6 months of captopril therapy seems to represent a transient phenomenon as UAER increases after 12 months to levels similar to the previous values. GFR and RPF decreased slightly or remained constant (a tendency to a more pronounced fall in GFR rate with placebo treatment was noted) thereby giving evidence for preservation of renal function over the observation period.

The majority of studies with type 1 or type 2 diabetic patients investigating the effect of ACE inhibitors in incipient nephropathy have demonstrated a reduction in UAER over various treatment periods, usually accompanied by arresting or slowing of the progression of renal disease. While captopril has been shown to produce a significant reduction in UAER in most studies over shorter treatment periods [1, 12, 23], the results of long-term studies are controversial [6, 21, 22, 32].

Marre et al. [19] reported that median albumin excretion in normotensive type 1 diabetic patients was reduced during enalapril treatment over an observation period of 6 weeks. At our first and second visits (after 3 and 6 months) a decline in UAER was observed (Figs. 2, 3) especially in the small hypertensive subgroup. Mimran et al. [23] observed type 1 diabetic patients with microalbuminuria treated with ACE inhibitors and CA channal blockers for 6 weeks and found a decrease in UAER of 40% during captopril and an increase in UAER of 40% with nifedipine treatment. Baba et al. [1] observed in hypertensive type 2 diabetic patients a significant reduction in UAER during enalapril and nicardipine without altering RPF and GFR after 4–6 weeks. Possibly, ACE inhibitors lower UAER in type 2 diabetes patients only during the first period of medication.

The Melbourne Diabetic Nephropathy Study Group [22] reported the results of a 12-month study comparing the efficacy of perindopril and nifedipine on microalbuminuria in normotensive and hypertensive type 1 and 2 diabetic patients, and observed a reduction of UAER in all hypertensive patients. Another long-term study (30 months) following the effectiveness of ACE inhibitor captopril in preventing development of diabetic nephropathy in normotensive type 1 diabetes patients was reported by Mathiesen et al. [21]. UAER was gradually reduced, but not significantly, whereas in the placebo group an increase occurred; this

suggests postponement in development of clinically overt diabetic nephropathy.

Similarly, Ravid et al. [32] showed in normotensive type 2 diabetic patients treated with enalapril the long-term stabilization of plasma creatinine levels and the degree of proteinuria after 5-year medication. In the placebo group the fifth-year average of UAER was 152% greater than the baseline values. Comparison of these data to our results is limited because of differences in follow-up time, age, the number of patients, and duration of diabetes. The divergence in UAER values between the two groups increased after the 2-year and subsequent follow-ups.

Chan et al. [6] compared the efficacy of enalapril and nifedipine in hypertensive type 2 diabetic patients. In the enalapril group proteinuria was reduced more significantly after 1-year therapy with enalapril than with nifedipine, but plasma creatinine levels increased in the enalapril group by 20%. The dose of the ACE inhibitor was four times higher than in the 5-year study of Ravid et al. [32]. In our study only a transient decrease in microalbuminuria was found; creatinine levels remained stable.

Most of these studies, however, were performed in patients with type 1 diabetes, which may explain the divergent results. Our results indicate that captopril leads to a transient decrease in microalbuminuria in hypertensive type 2 diabetic patients, which reverses to baseline values after 12 months. Renal function, however, seems to be constant, as demonstrated by the lack of significant alteration in GFR and RPF. More recent studies of various ACE inhibitors such as enalapril [1, 2, 6, 32, 37] and perindopril [22] exhibited beneficial effects on UAER. This could be explained by different tissue-specific effects of these ACE inhibitors compared to captopril.

Finally, the influence of antihypertensive therapy on kidney function is not definitively enlightened in type 2 diabetic patients [15]. Different mechanisms are supposed. Lowering systemic arterial blood pressure leads to a decrease in glomerular perfusion pressure and thus to a reduction in efferent arteriolar resistance; therefore UAER decreases [38]. Other possibilities include reduced glomerular pore size during enalapril medication [26]. Whether treatment with ACE inhibitors does in fact reduce the glomerular lesions in diabetic nephropathy remains to be established.

In conclusion, the results of this study confirm the short-term effect of captopril in lowering albumin excretion in type 2 diabetic patients with microalbuminuria, whereas beneficial effects of long-term medication remain to be established. GFR does not decline (during captopril treatment), thereby suggesting constant renal function. Whether ACE inhibitors are able to reverse increased UAER and maintain renal function over a longer period needs further evaluation in type 2 diabetes mellitus.

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